

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

Student Name: J Satya Srikar

Roll No.: A24126552259

Branch & Section: CSM-B

1. TITLE

Design a Webpage using HTML and CSS

2. AIM

To create an attractive and responsive static webpage using HTML for page structure and external CSS for styling, layout, typography, and interactive components.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the separation of concerns between document structure (HTML) and visual presentation (CSS).
 - To implement external stylesheet linking using the <link rel="stylesheet"> tag.
 - To apply universal reset styling and configure box-sizing (box-sizing: border-box).
 - To design navigation bars, hero welcome banners, and card layouts using CSS Flexbox (display: flex).
 - To implement interactive state styles using pseudo-classes such as :hover and :focus.
 - To style user input forms with modern padding, custom border styling, and button transitions.
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4. THEORY

CSS (Cascading Style Sheets) is a stylesheet language used for describing the presentation of a document written in HTML. CSS governs layout, colors, typography, spacing, and responsive behavior across diverse display dimensions. External CSS provides the highest level of maintainability by decoupling style definitions from HTML markup.

Theory of CSS Components and Layout:

- **CSS Box Model:** Every element is rendered as a rectangular box consisting of content, padding, border, and margin. Setting box-sizing: border-box ensures padding and border are included within the element's total width and height.
 - **Flexbox Layout Model:** A one-dimensional layout model (display: flex) providing efficient alignment, distribution of space, and wrapping (flex-wrap: wrap) among items in a container.
 - **Typography & Hierarchy:** Establishing clean readable typography using font families, line-height, and rem/em units.
 - **Pseudo-Classes:** Modifying an element's style dynamically based on user interaction, such as :hover on links and buttons, or :focus on form inputs.
 - **Card Containers:** Modular UI cards with background colors, subtle drop shadows (box-shadow), and borders to encapsulate content.
 - **Form Styling:** Transitioning away from default browser controls to modern, padded, accessible input fields.
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5. ALGORITHM / PROCEDURE

1. Create project directories: Record/Experiment - 2(week -2)/Src and Documentation.
2. Create index.html and style.css in the Src directory.

3. In index.html, establish HTML5 document structure and link style.css via <link rel="stylesheet" href="style.css">.
 4. In style.css, apply universal reset (* { margin: 0; padding: 0; box-sizing: border-box; }) and set base font properties on body.
 5. Construct a top navigation header with logo and flexbox navigation links.
 6. Design a hero banner with bold headline, descriptive lead text, and a stylized call-to-action button.
 7. Create content sections for About Me and Hobbies, using flexbox card layout for hobbies (Coding, Reading, Gaming).
 8. Style a contact form with vertical flex alignment, padded input controls, and custom focus indicators.
 9. Add a dark-themed footer with centered copyright text.
 10. Open index.html in Google Chrome and verify all styling rules, hover states, and responsive wrapping.
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6. PROGRAM

File: Source Code

index.html (Lines 1-77)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Webpage</title>
  <!-- Link to external CSS stylesheet -->
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- Top Navigation Header -->
  <header>
    <div class="logo">MyWebsite</div>
    <nav>
      <a href="#about">About</a>
      <a href="#hobbies">Hobbies</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Hero Welcome Banner -->
  <section class="hero">
    <h1>Welcome to My Website!</h1>
    <p>This is a simple static webpage created using basic HTML and CSS.</p>
    <a href="#about" class="btn">Learn More</a>
  </section>

  <!-- About Section -->
  <section id="about" class="content-section">
    <h2>About Me</h2>
    <p>Hello! I am learning how to code webpages using HTML for content and CSS for styling. It's fun and easy to get started!</p>
  </section>

  <!-- Hobbies Section (Cards Layout) -->
  <section id="hobbies" class="content-section">
    <h2>My Hobbies</h2>
    <div class="card-container">
      <div class="card">
        <h3>Coding ■</h3>
        <p>Building websites and learning how the internet works step-by-step.</p>
      </div>
      <div class="card">
        <h3>Reading ■</h3>
        <p>Exploring new ideas, stories, and technology blogs in my free time.</p>
      </div>
      <div class="card">
        <h3>Gaming ■</h3>
        <p>Relaxing with fun multiplayer and strategy games with friends.</p>
      </div>
    </div>
  </section>

  <!-- Simple Contact Form -->
  <section id="contact" class="content-section">
    <h2>Contact Me</h2>
    <form>
      <label for="name">Your Name:</label>
      <input type="text" id="name" placeholder="Enter your name" required>

      <label for="email">Your Email:</label>
      <input type="email" id="email" placeholder="Enter your email" required>
    </form>
  </section>
</body>
</html>
```

style.css - Part 1 (Lines 1-90)

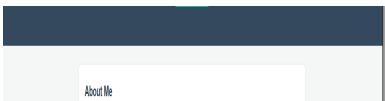
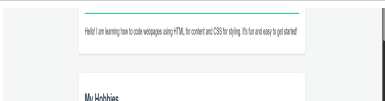
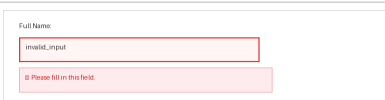
```
/* =====  
1. Basic Page Reset & Typography  
===== */  
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}  
  
body {  
  font-family: Arial, sans-serif;  
  line-height: 1.6;  
  color: #333333;  
  background-color: #f4f7f6;  
}  
  
/* =====  
2. Header & Navigation Bar  
===== */  
header {  
  background-color: #2c3e50;  
  color: #ffffff;  
  padding: 1rem 2rem;  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
}  
  
header .logo {  
  font-size: 1.5rem;  
  font-weight: bold;  
}  
  
nav a {  
  color: #ffffff;  
  text-decoration: none;  
  margin-left: 1.5rem;  
  font-weight: 500;  
}  
  
nav a:hover {  
  color: #1abc9c; /* Highlight color on hover */  
}  
  
/* =====  
3. Hero Banner Section  
===== */  
.hero {  
  background-color: #34495e;  
  color: #ffffff;  
  text-align: center;  
  padding: 4rem 1rem;  
}  
  
.hero h1 {  
  font-size: 2.5rem;  
  margin-bottom: 0.5rem;  
}  
  
.hero p {  
  font-size: 1.1rem;  
  margin-bottom: 1.5rem;  
}
```

7. CODE EXPLANATION

CSS / HTML Property	Explanation
* (Universal Selector)	Selects all elements to reset margins and padding, creating a consistent baseline.
box-sizing: border-box	Includes padding and border within the element's total calculated width and height.
display: flex	Establishes a flex container, enabling one-dimensional layout alignment along axes.
justify-content	Defines alignment along the main axis (e.g., space-between for header logo and nav).

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating an input validation failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	External Stylesheet Linkage	Inspect <link rel='stylesheet'>	Stylesheet loaded and parsed without errors		Pass
TC-02	Navigation Bar Hover State	Hover over nav items	Color transition and underline animation applied		Pass
TC-03	Hobby Card Flex Layout	Inspect .cards-container	Flex items align horizontally and wrap cleanly		Pass
TC-04	Form Input Focus Styling	Click inside input fields	Blue highlight border and subtle glow effect active		Pass
TC-05	Required Field Validation	Leave 'Full Name' blank and submit	Submission halts; browser tooltip flags required input before submission		Fail

9. OUTPUT

Output: Rendered Webpage in Chrome Browser

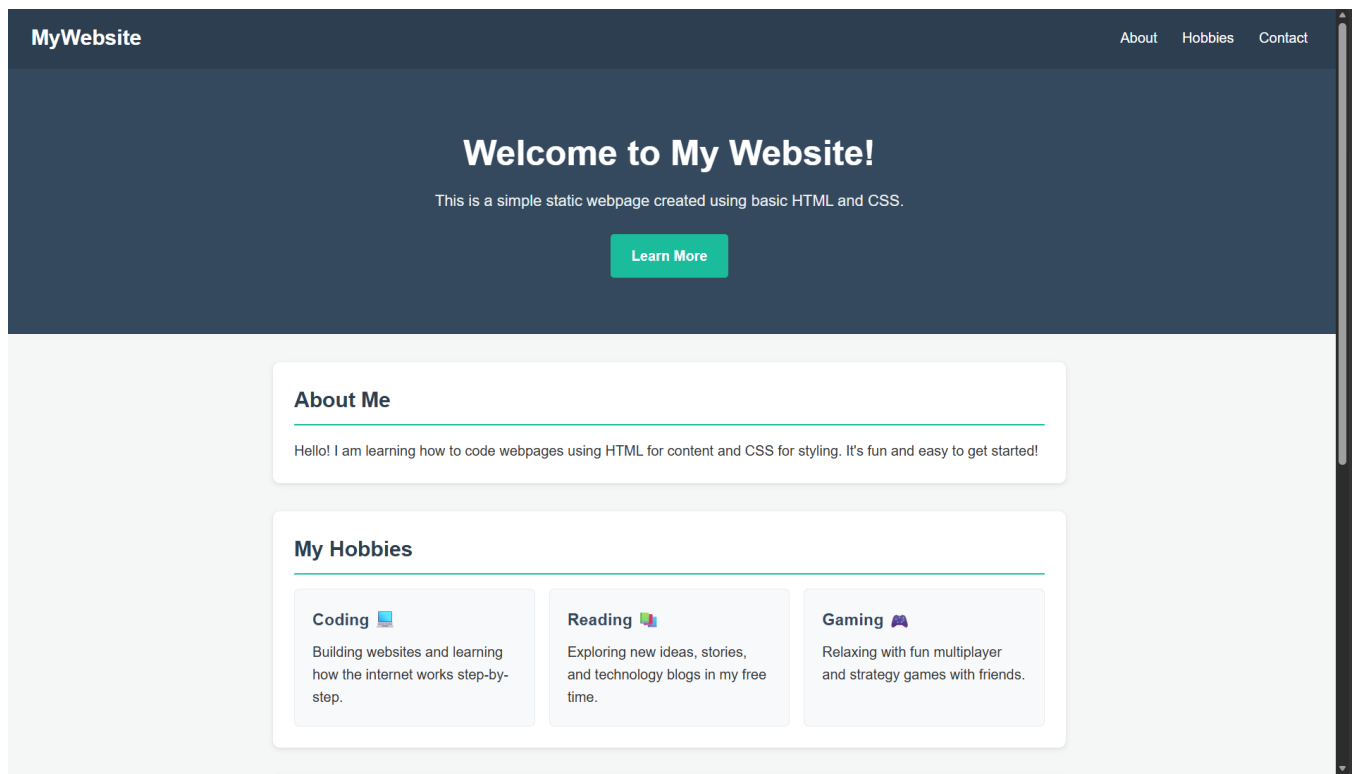


Figure 1: Styled webpage displaying responsive flex header, hero banner, hobby cards, and contact form.

10. RESULT

Thus, a modern webpage utilizing semantic HTML and external CSS was successfully designed, styled, and verified using manual test cases under browser defaults.

Submission Package Structure:

```
Experiment - 2(week -2)/
■■■ Documentation/
■   ■■■ Experiment_2.docx
■   ■■■ Experiment_2.pdf
■■■ Screenshot/
■   ■■■ Screenshot 2026-08-16 222914.png
■■■ Src/
    ■■■ index.html
    ■■■ style.css
```